Nonlinear Functions, Function Notation & Equivalent Expressions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the municipal library is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 4$, what is the maximum value of $g(x)$?

Answer: 7. Reading the graph of $y = f(x)$, the maximum point (vertex) is labeled at $(0, 3)$, so the maximum value of $f(x)$ is 3. Since $g(x) = f(x) + 4$ shifts the graph vertically upward by 4 units, the maximum value of $g(x)$ is $3 + 4 = 7$.

2 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the coastal aquarium is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 7$, what is the maximum value of $g(x)$?

Answer: 13. Reading the graph of $y = f(x)$, the maximum point (vertex) is labeled at $(-1, 6)$, so the maximum value of $f(x)$ is 6. Since $g(x) = f(x) + 7$ shifts the graph vertically upward by 7 units, the maximum value of $g(x)$ is $6 + 7 = 13$.

3 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the urban bike-share station is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 4$, what is the maximum value of $g(x)$?

Answer: 8. Reading the graph of $y = f(x)$, the maximum point (vertex) is labeled at $(2, 4)$, so the maximum value of $f(x)$ is 4. Since $g(x) = f(x) + 4$ shifts the graph vertically upward by 4 units, the maximum value of $g(x)$ is $4 + 4 = 8$.

4 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the agricultural cooperative is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 7$, what is the maximum value of $g(x)$?

Answer: 9. Reading the graph of $y = f(x)$, the maximum point (vertex) is labeled at $(1, 2)$, so the maximum value of $f(x)$ is 2. Since $g(x) = f(x) + 7$ shifts the graph vertically upward by 7 units, the maximum value of $g(x)$ is $2 + 7 = 9$.

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5 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the recycling center is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 4$, what is the maximum value of $g(x)$?

Answer: 9. Reading the graph of $y = f(x)$, the maximum point (vertex) is labeled at $(0, 5)$, so the maximum value of $f(x)$ is 5. Since $g(x) = f(x) + 4$ shifts the graph vertically upward by 4 units, the maximum value of $g(x)$ is $5 + 4 = 9$.

6 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the neighborhood bookstore is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 7$, what is the maximum value of $g(x)$?

Answer: 10. Reading the graph of $y = f(x)$, the maximum point (vertex) is labeled at $(-1, 3)$, so the maximum value of $f(x)$ is 3. Since $g(x) = f(x) + 7$ shifts the graph vertically upward by 7 units, the maximum value of $g(x)$ is $3 + 7 = 10$.

7 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the culinary institute is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 4$, what is the maximum value of $g(x)$?

Answer: 10. Reading the graph of $y = f(x)$, the maximum point (vertex) is labeled at $(2, 6)$, so the maximum value of $f(x)$ is 6. Since $g(x) = f(x) + 4$ shifts the graph vertically upward by 4 units, the maximum value of $g(x)$ is $6 + 4 = 10$.

Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the robotics club, in a calculation for the orchestra hall (case 12), if $f(x) = 2|x + 9| - 1$, what is $f(10)$?

Answer: 37. Applying Nonlinear Functions & Function Notation principles, $f(10) = 2|10 + 9| - 1 = 2(19) - 1 = 37$ Therefore, the verified correct option is 37.

9 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the maple syrup farm, for what value of x is the function $f(x) = \frac{1}{x - 5}$ undefined?

Answer: 5. The denominator is zero when $x - 5 = 0 \Rightarrow x = 5$.

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10 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the marine research vessel, in a calculation for the local farmers market (case 20), what is the minimum value of $f(x) = 1|x - 7| - 8$?

Answer: -8. Since $1|x - 7| \geq 0$ always, the minimum occurs at $x = 7$, giving $f(7) = -8$.

11 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the campus coffee roastery, in a calculation for the performing arts academy (case 24), if $f(7) = 17$, and $g(x) = f(x) - 1$, what is $g(7)$?

Answer: 16. Applying Nonlinear Functions & Function Notation principles, $g(7) = f(7) - 1 = 17 - 1 = 16$ Therefore, the verified correct option is 16.

12 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the meteorological station, in a calculation for the coastal aquarium (case 28), if $f(-8) = -7$, and $g(x) = f(x) + 4$, what is $g(-8)$?

Answer: -3. Applying Nonlinear Functions & Function Notation principles, $g(-8) = f(-8) + 4 = -7 + 4 = -3$ Therefore, the verified correct option is -3.

13 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the urban bike-share station, in a calculation for the public transit depot (case 32), if $f(3) = -12$, and $g(x) = f(x) - 2$, what is $g(3)$?

Answer: -14. Applying Nonlinear Functions & Function Notation principles, $g(3) = f(3) - 2 = -12 - 2 = -14$ Therefore, the verified correct option is -14.

14 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the sports complex, in a calculation for the geology lab (case 40), if $h(x) = |x - 4| + 2$, what is the minimum value of $h(x)$?

Answer: 2. Since $|x - 4| \geq 0$ with equality at $x = 4$, the minimum value of $h(x)$ is $0 + 2 = 2$.

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the regional airport shuttle, what is the smallest value of x for which $f(x) = \sqrt{x}$ is defined?

Answer: 0. We need $x \geq 0 \Rightarrow x \geq 0$. The smallest such value is 0.

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16 [Challenging] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

At the botanical conservatory, the functions f and g are defined by $f(x) = 2x - 2$ and $g(x) = x^2 + 1$. What is the value of $f(g(2))$?

Answer: 8. First evaluate the inner function: $g(2) = (2)^2 + 1 = 5$. Then substitute into f: $f(5) = 2(5) - 2 = 8$.

17 [Challenging] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

In the xy-plane, the graph of the quadratic function $y = f(x)$ has its vertex at $(3, 4)$. If $g(x) = f(x - 5)$ models the shifted curve at the municipal library, what is the x-coordinate of the vertex of the graph of g?

Answer: 8. The transformation $g(x) = f(x - 5)$ shifts the graph of f horizontally to the right by 5 units. Therefore, the x-coordinate of the vertex moves from 3 to $3 + 5 = 8$.

18 [Challenging] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

The nonlinear function $h(x) = 4x^2 + c$, where c is a constant, models the output at the community theater. If $h(4) = 71$, what is the value of $h(0)$?

Answer: 7. Substitute $x = 4$ into $h(x)$: $4(4)^2 + c = 71 \Rightarrow 64 + c = 71 \Rightarrow c = 7$. Since $h(0) = c$, we have $h(0) = 7$.

19 [Challenging] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

At the regional bakery, the functions f and g are defined by $f(x) = 5x - 5$ and $g(x) = x^2 + 4$. What is the value of $f(g(5))$?

Answer: 140. First evaluate the inner function: $g(5) = (5)^2 + 4 = 29$. Then substitute into f: $f(29) = 5(29) - 5 = 140$.

20 [Challenging] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

In the xy-plane, the graph of the quadratic function $y = f(x)$ has its vertex at $(6, 7)$. If $g(x) = f(x - 8)$ models the shifted curve at the coastal aquarium, what is the x-coordinate of the vertex of the graph of g?

Answer: 14. The transformation $g(x) = f(x - 8)$ shifts the graph of f horizontally to the right by 8 units. Therefore, the x-coordinate of the vertex moves from 6 to $6 + 8 = 14$.

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

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21 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the botanical conservatory is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 3$, what is the maximum value of $g(x)$?

a. 5 ✓

b. 2

c. -1

d. 1

Reading the graph of $y = f(x)$, the maximum point (vertex) is labeled at $(-1, 2)$, so the maximum value of $f(x)$ is 2. Since $g(x) = f(x) + 3$ shifts the graph vertically upward by 3 units, the maximum value of $g(x)$ is $2 + 3 = 5$.

22 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the community theater is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 5$, what is the maximum value of $g(x)$?

a. -1

b. 5

c. 9 ✓

d. 6

Reading the graph of $y = f(x)$, the maximum point (vertex) is labeled at $(1, 4)$, so the maximum value of $f(x)$ is 4. Since $g(x) = f(x) + 5$ shifts the graph vertically upward by 5 units, the maximum value of $g(x)$ is $4 + 5 = 9$.

23 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the regional bakery is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 6$, what is the maximum value of $g(x)$?

a. 7

b. 11 ✓

c. 8

d. -1

Reading the graph of $y = f(x)$, the maximum point (vertex) is labeled at $(2, 5)$, so the maximum value of $f(x)$ is 5. Since $g(x) = f(x) + 6$ shifts the graph vertically upward by 6 units, the maximum value of $g(x)$ is $5 + 6 = 11$.

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24 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the mountain observatory is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 8$, what is the maximum value of $g(x)$?

a. 8

b. -6

c. 6

d. 10 ✓

Reading the graph of $y = f(x)$, the maximum point (vertex) is labeled at $(0, 2)$, so the maximum value of $f(x)$ is 2. Since $g(x) = f(x) + 8$ shifts the graph vertically upward by 8 units, the maximum value of $g(x)$ is $2 + 8 = 10$.

25 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the science museum is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 3$, what is the maximum value of $g(x)$?

a. 0

b. 2

c. 6 ✓

d. 4

Reading the graph of $y = f(x)$, the maximum point (vertex) is labeled at $(1, 3)$, so the maximum value of $f(x)$ is 3. Since $g(x) = f(x) + 3$ shifts the graph vertically upward by 3 units, the maximum value of $g(x)$ is $3 + 3 = 6$.

26 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the solar microgrid facility is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 5$, what is the maximum value of $g(x)$?

a. 10 ✓

b. 4

c. 0

d. 6

Reading the graph of $y = f(x)$, the maximum point (vertex) is labeled at $(-1, 5)$, so the maximum value of $f(x)$ is 5. Since $g(x) = f(x) + 5$ shifts the graph vertically upward by 5 units, the maximum value of $g(x)$ is $5 + 5 = 10$.

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27 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the ceramic studio is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 6$, what is the maximum value of $g(x)$?

a. 6

b. 0

c. 8

d. 12 ✓

Reading the graph of $y = f(x)$, the maximum point (vertex) is labeled at $(0, 6)$, so the maximum value of $f(x)$ is 6. Since $g(x) = f(x) + 6$ shifts the graph vertically upward by 6 units, the maximum value of $g(x)$ is $6 + 6 = 12$.

28 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the publishing house is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 8$, what is the maximum value of $g(x)$?

a. 7

b. 11 ✓

c. 10

d. -5

Reading the graph of $y = f(x)$, the maximum point (vertex) is labeled at $(2, 3)$, so the maximum value of $f(x)$ is 3. Since $g(x) = f(x) + 8$ shifts the graph vertically upward by 8 units, the maximum value of $g(x)$ is $3 + 8 = 11$.

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

29 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the art gallery, in a calculation for the wildlife sanctuary (case 11), if $f(x) = 1|x + 7| - 8$, what is $f(-10)$?

a. -9

b. -5 ✓

c. -6

d. 4

Applying Nonlinear Functions & Function Notation principles,
$f(-10) = 1| - 10 + 7| - 8 = 1(3) - 8 = -5$ Therefore, the verified correct option is -5.

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30 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the culinary institute, in the coordinate plane, the graph of $y = F(w)$ is translated 6 units right and 5 units up to form $y = G(w)$, mapping $D(12, 33)$ to E. What is the sum of coordinates of E on $y = G(w)$?

a. 56 ✓

b. 50

c. 61

d. 63

The image point is $E(18, 38)$, whose coordinates sum to 56.

31 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the water treatment plant, in a calculation for the textile workshop (case 14), what is the minimum value of $f(x) = 1|x - 9|$?

a. -3

b. 0 ✓

c. -6

d. 6

Since $1|x - 9| \geq 0$ always, the minimum occurs at $x = 9$, giving $f(9) = 0$.

32 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the sports complex, in a calculation for the geology lab (case 15), what is the minimum value of $f(x) = 7|x + 5| - 10$?

a. -16

b. -8

c. -5

d. -10 ✓

Since $7|x + 5| \geq 0$ always, the minimum occurs at $x = -5$, giving $f(-5) = -10$.

33 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the planetarium, in a calculation for the carpentry workshop (case 17), if $f(-7) = -18$, and $g(x) = f(x) - 3$, what is $g(-7)$?

a. -21 ✓

b. -12

c. -25

d. -16

Applying Nonlinear Functions & Function Notation principles, $g(-7) = f(-7) - 3 = -18 - 3 = -21$ Therefore, the verified correct option is -21.

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34 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the regional airport shuttle, in a calculation for the historic lighthouse (case 18), if $f(8) = 11$, and $g(x) = f(x) + 5$, what is $g(8)$?

a. 8

b. 22

c. 16 ✓

d. 10

Applying Nonlinear Functions & Function Notation principles, $g(8) = f(8) + 5 = 11 + 5 = 16$ Therefore, the verified correct option is 16.

35 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the wind energy cooperative, consider the quadratic function $F(w) = 3w^2 - 15$ over the closed interval $[13, 18]$. What is the average rate of change $\Delta F/\Delta w$ on $[13, 18]$?

a. 96

b. 106

c. 93 ✓

d. 78

The average rate of change is $\frac{F(18) - F(13)}{5} = 3((18) + 13) = 93$.

36 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the national park visitor center, a function F satisfies the identity $F(w - 26) = 11w + 6$ for all real w. What is the value $F(13)$?

a. 435 ✓

b. 441

c. 446

d. 409

Setting $w - 26 = 13 \Rightarrow w = 39$ gives $F(13) = 11(39) + 6 = 435$.

37 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the glassblowing studio, in a calculation for the neighborhood hardware store (case 22), what is the minimum value of $f(x) = 6|x + 7| + 3$?

a. -1

b. 11

c. 3 ✓

d. -4

Since $6|x + 7| \geq 0$ always, the minimum occurs at $x = -7$, giving $f(-7) = 3$.

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Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the maple syrup farm, let the functions R and S be defined by $R(m) = 7m - 9$ and $S(m) = m^2 + 12$. What is the composite value $R(S(9))$?

a. 642 ✓

b. 654

c. 633

d. 649

First compute $S(9) = 9^2 + 12 = 93$. Then $R(93) = 7(93) - 9 = 642$.

39 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the planetarium, $f(x) = \{-4x - 8, x < -5;\ -4x + 7, x \geq -5$ What is $f(-9)$?

a. 26

b. 28 ✓

c. 31

d. 21

Since $-9 < -5$, use the first piece: $f(-9) = 28$.

40 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the wind energy cooperative, in a calculation for the community orchard (case 69), if $f(x) = 4x^2 + 6$ and $g(x) = 4x - 8$, what is $f(g(1))$?

a. 61

b. 77

c. 70 ✓

d. 74

Applying Nonlinear Functions & Function Notation principles, $g(1) = -4$. $f(-4) = 4(-4)^2 + 6 = 70$. Therefore, the verified correct option is 70.

41 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the marine research vessel, in the coordinate plane, the graph of $y = R(m)$ is translated 5 units right and 3 units up to form $y = S(m)$, mapping $M(17, 15)$ to N. What is the sum of coordinates of N on $y = S(m)$?

a. 35

b. 40 ✓

c. 43

d. 51

The image point is $N(22, 18)$, whose coordinates sum to 40.

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42 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

The nonlinear function $h(x) = 5x^2 + c$, where c is a constant, models the output at the robotics club. If $h(3) = 67$, what is the value of $h(0)$?

a. 67

b. 27

c. 19

d. 22 ✓

Substitute $x = 3$ into $h(x)$: $5(3)^2 + c = 67 \Rightarrow 45 + c = 67 \Rightarrow c = 22$. Since $h(0) = c$, we have $h(0) = 22$.

43 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

At the university lecture hall, the functions f and g are defined by $f(x) = 5x - 6$ and $g(x) = x^2 + 1$. What is the value of $f(g(4))$?

a. 15

b. 85

c. 79 ✓

d. 17

First evaluate the inner function: $g(4) = (4)^2 + 1 = 17$. Then substitute into f: $f(17) = 5(17) - 6 = 79$.

44 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

In the xy-plane, the graph of the quadratic function $y = f(x)$ has its vertex at $(3, 8)$. If $g(x) = f(x - 8)$ models the shifted curve at the culinary institute, what is the x-coordinate of the vertex of the graph of g?

a. 3

b. 11 ✓

c. -5

d. 16

The transformation $g(x) = f(x - 8)$ shifts the graph of f horizontally to the right by 8 units. Therefore, the x-coordinate of the vertex moves from 3 to $3 + 8 = 11$.

45 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

The nonlinear function $h(x) = 8x^2 + c$, where c is a constant, models the output at the wildlife sanctuary. If $h(2) = 57$, what is the value of $h(0)$?

a. 25 ✓

b. 57

c. 33

d. 23

Substitute $x = 2$ into $h(x)$: $8(2)^2 + c = 57 \Rightarrow 32 + c = 57 \Rightarrow c = 25$. Since $h(0) = c$, we have $h(0) = 25$.

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